b- $\int_0^{+\infty} x^3 e^{-x^2}\,dx$

$\int_0^{+\infty} e^{-x}\,dx = \left[-e^{-x}\right]_0^{\infty} = 0 - -e^0 = 1$

$\int_0^{\Pi} x^3 e^{-x^2}\,dx$ $\quad \frac{d}{dx}\left(e^{-x^2}\right) = -2x \cdot e^{-x^2}$

$\int x^3 e^{-x^2} = \int \underbrace{\left(\frac{-1}{2}x^2\right)}_{u} \cdot \underbrace{(-2x)e^{-x^2}}_{v'} = \underbrace{\left[\frac{-1}{2}x^2 \cdot e^{-x^2}\right]_0^{\infty}}_{x\to+\infty \quad \frac{u}{e^u} \to 0} - \int_0^{\infty}(-x)e^{-x^2}\,dx = +\int_0^{\infty}\left(\frac{1}{2}\right)(-2)xe^{-x^2}\,dx$

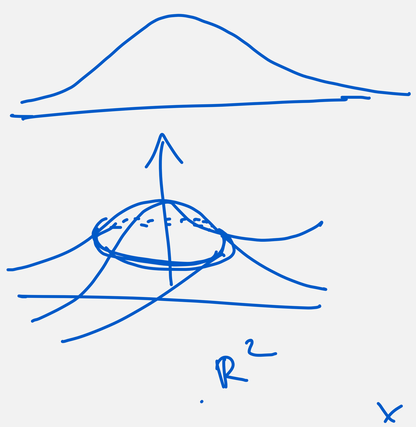
$= -\frac{1}{2}\int_0^{\infty} -2x e^{-x^2}\,dx = \frac{-1}{2}\left[e^{-x^2}\right]_0^{\infty} = \frac{+1}{2}\cdot 1$

$y \to \int_0^{y} \underbrace{e^{-x^2}}\,dx$ $\quad = 0$

ni poly, ni cos/sin, ni log, exp, ni a fractions

ni en $\times, +, -, \div$, compo

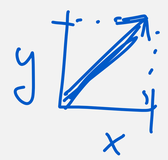
$\int_{-\infty}^{+\infty} e^{-x^2}\,dx = \sqrt{2\pi}$



$\iint_{\mathbb{R}\,\mathbb{R}} e^{-x^2} e^{-y^2}\,dx\,dy = \left(\int e^{-x^2}\right)^2$

$x^3 \quad e^{-x^2}$

$(x,y), \quad x^2+y^2$



$e^x = \sum \frac{x^n}{n!}$ $\qquad \int \sum = \sum \int$